6.7 Measuring, time and money: answers

← Unit overview · Practice

For parents: for time questions, accept equivalent answers (3:30 or 15:30 for item 4.1; “1 hour 25 minutes” or “85 minutes”). Let him check measuring answers with a real ruler, jug or clock.

Set 1 · Warm-up

#	Answer	How
1	100	
2	1 000	*Kilo* means a thousand.
3	1 000	*Milli* means a thousandth.
4	60	
5	60	
6	30	Half of 60.
7	1 000	*Kilo* again.
8	7	
9	€7	
10	15	A quarter of 60.

Set 2 · Length, weight and capacity

#	Answer	How
1	300	3 × 100.
2	2 m and 50 cm	200 cm = 2 m, 50 cm left. (Also 2,5 m.)
3	4 000	4 × 1 000.
4	1,5 km (or 1 km and 500 m)	1 000 m = 1 km, 500 m = half a km.
5	2 000	2 × 1 000.
6	1 kg	750 + 250 = 1 000 g.
7	10	
8	2 000	2 × 1 000.
9	4 glasses	4 × 250 ml = 1 000 ml.
10	50	5 × 10.
11	10	

Set 3 · Perimeter and area

#	Answer	How
1	16 cm	5 + 3 + 5 + 3.
2	15 cm²	3 rows of 5 squares.
3	16 cm	4 + 4 + 4 + 4.
4	16 cm²	4 rows of 4.

#	Answer	How
5	Shape B	16 cm² is more than 15 cm².
6	16 m	6 + 2 + 6 + 2.
7	12 m²	2 rows of 6 square metres.
8	4 squares	6 × 4 = 24.
9	40 cm	4 sides of 10 cm.

Talk about it: same perimeter, different area. A square-ish shape gets more area from the same fence.

Set 4 · Telling the time

#	Answer	How
1	3:30 (or 15:30)	Long hand at 6 is “half”; short hand between 3 and 4. In Dutch: *half vier*.
2	20:15	Evening, so add 12 to the hour: 8 + 12 = 20.
3	6:45 (06:45)	A quarter before seven.
4	16:20	*Half vijf* = 4:30; ten minutes before that is 4:20, in the afternoon 16:20.
5	*half drie (’s middags)*	2:30 in the afternoon. “Half” points to the next hour, three.
6	*tien voor acht (’s avonds)*	7:50 in the evening.
7	1 hour 25 minutes	18:45 → 19:00 is 15 min, → 20:00 is 1 hour, → 20:10 is 10 min.
8	45 minutes	10:47 → 11:00 is 13 min, → 11:32 is 32 min. 13 + 32 = 45.
9	120	2 × 60.
10	8:10	20 minutes before 8:30.
11	75	60 + 15.

Set 5 · Calendar and money

#	Answer	How
1	365	(A leap year has 366.)
2	30	“30 days has September, April, June and November.” Or the knuckle trick.
3	13 days	14 − 1.

#	Answer	How
4	Thursday	7 days later is Monday again, then 3 more days.
5	14	2 × 7.
6	€2 + €1 + 50 cent + 20 cent + 10 cent + 5 cent (6 coins)	€3 with two coins, 85 cents with four.
7	€6,60	Count up: €13,40 → €14 is 60 cents, → €20 is €6.
8	€6,75	3 × €2 = €6, 3 × 25 cents = 75 cents.
9	About €7; exactly €7,23	Estimate €1 + €2 + €4 = €7. Exact: €1,29 + €2,15 + €3,79.
10	€0,50 (50 cents)	

Set 6 · Mixed word problems

#	Answer	How
1	10:15	9:48 → 10:00 is 12 min; 27 − 12 = 15 more minutes.
2	2 cartons	1,5 l = 1 500 ml = 2 × 750 ml.
3	25 km	12,5 + 12,5.
4	4 kg 200 g	2 300 g + 1 900 g = 4 200 g.
5a	26 m	8 + 5 + 8 + 5.
5b	40 m²	8 × 5.
6	5 hours 45 minutes	8:30 → 14:30 is 6 hours; 14:15 is 15 minutes less.
7	180 minutes = 3 hours	4 × 45 = 180; 180 : 60 = 3.
8	€1,20	500 g is half a kilo; half of €2,40.
9	24	

Puzzle corner

With 24 cm of string, the two sides next to each other always add up to **12 cm** (half of 24). The rectangles with whole-centimetre sides are:

Sides	Area
1 × 11	11 cm²
2 × 10	20 cm²
3 × 9	27 cm²
4 × 8	32 cm²
5 × 7	35 cm²
6 × 6	36 cm²

Biggest area: the **square**, 6 × 6 = 36 cm². Smallest: the long thin 1 × 11 = 11 cm². The pattern: the closer to a square, the bigger the area. Spotting “the sides always add up to 12” is already a big insight.